# Md Aquib Molla

Curriculum Vitae

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AquibMolla

## Research Profile

I work in non-equilibrium statistical physics, focusing on stochastic processes in spatially extended systems. My research investigates how geometry, motion, and resource constraints interact to produce emergent transport, survival behavior, and collective dynamics. I develop minimal theoretical models combining analytical approaches and numerical simulations.

Recently, I have started working on active particle models involving stochastic dynamics and memory effects, including agent-based simulations and extensions toward kinetic theory. Through this work, I have become increasingly interested in active matter, collective behavior, and the emergence of nonequilibrium phenomena from local interactions.

## Education

2021–present **Ph.D. in Physics**, *Vidyasagar College, University of Calcutta*, *Expected: July 2026*

2018–2020 **M.Sc. in Physics**, *University of Calcutta*, *CGPA: 7.15*

2015–2018 **B.Sc. in Physics**, *University of Calcutta*

## Research Interests

- Active matter and collective dynamics
- Nonequilibrium stochastic processes (random walks, first-passage phenomena)
- Data assimilation, inverse problems, and identification of minimal dynamical models from time-series data
- Adaptive and interacting particle systems (foraging dynamics, resource feedback)
- Geometry and disorder (percolation, connectivity effects)

## Publications

2026 **M. A. Molla, S. Goswami, P. Sen**, *Forager with intermittent rest: Better for survival?*, *Physical Review E*, **113**, 034139

2026 **P. Maji, M. A. Molla, K. Dey, et al.**, *Percolative Instabilities and Sparse-Limit Fractality in* $1T-TaS_2$, *Physical Review B*, **113**, 115110

2026 **M. A. Molla, S. Goswami**, *Energy Dynamics and Partial Consumption in Foraging*, accepted in Physica A

2024 **M. A. Molla, S. Goswami**, *An Insight into Heart-like Systems with Percolation*, *Physics Letters A*, **518**, 129695

2023 **M. A. Molla, S. Goswami**, *Quantum Walker in Presence of a Moving Detector*, *Physica A*, **620**, 128775

2026 **M. A. Molla, S. Goswami**, *Moving Detector Quantum Walk with Random Relocation*, arXiv:2604.03593

## Academic Visit

2025 **University of Potsdam, Germany**, Hosted by Prof. Ralf Metzler

## Selected Talks and Posters

2025 **Talk**, *University of Potsdam, **Germany***, From Cardiac Signals to Wandering Foragers

2025 **Talk**, *University of Trento, **Italy** (Online)*, Percolation in Excitable Media: Insights into Signal Propagation in Heart-like Systems

2025 **Poster**, *Driven and Active Amorphous Matter, Göttingen, **Germany***, Inertia-Driven Foraging

2025 **Talk**, *Leuven School on Nonequilibrium Statistical Mechanics, KU Leuven, **Belgium***, Cardiac Dynamics through Percolation

2025 **Poster**, *Decisions, Games and Evolution, ICTS Bengaluru, India*, Strategic Foraging

2024 **Talk**, *Discrete Integrable Systems, ICTS Bengaluru, India*, Percolation in Cardiac Systems

## Selected Workshops and Schools

2025 Bangalore School on Statistical Physics (ICTS)

2025 Driven and Active Amorphous Matter, Göttingen (Germany)

2025 Leuven School on Nonequilibrium Statistical Mechanics (Belgium)

2024 First-Passage Percolation and Related Models (ICTS)

## Honors

2019 CSIR–UGC NET (JRF)

2020 GATE Qualified

## Referees

**Dr. Sanchari Goswami**, Department of Physics, Vidyasagar College, University of Calcutta (sg.phys.caluniv@gmail.com)

**Prof. Ralf Metzler**, Institute of Physics and Astronomy, University of Potsdam (rmetzler@uni-potsdam.de)

## Computation & Technical Skills

C, Fortran, Python, Julia

LaTeX, Mathematica, MATLAB